Smart Door Lock System

Ojas Gaur (Leader)

Reg: 25BCE2388

Email:
ojas.gaur2025@
vitstudent.ac.in

Mob: +91 7525826839

Sumit Kumar

Reg: 25BCE0280

Email:
sumit.kumar2025b@
vitstudent.ac.in

Mob: +91 9334324355

Rushi Raj Meher

Reg: 25BCE2387

Email:
rushi.rajmeher2025@
vitstudent.ac.in

Mob: +91 8197037510

Problem Statement

Background & Core Problem

Conventional mechanical lock mechanisms and basic electronic keypads suffer from significant security, usability, and reliability limitations. Physical keys are easily lost, stolen, or duplicated, while standard keypad systems frequently rely on volatile memory, leading to credential loss during unexpected power resets. Furthermore, human errors—such as accidentally leaving doors unlocked or failing to close them completely—leave premises vulnerable to unauthorized intrusion. Standard standalone access solutions lack proactive brute-force mitigation, local status display feedback, and real-time door-state alarms required for secure, high-assurance facility monitoring.

Proposed Engineering Solution

To resolve these vulnerabilities, this project implements an embedded IoT access control system powered by an ESP32 microcontroller. The solution integrates dynamic 4-to-8 digit PIN authentication backed by non-volatile flash storage (‘Preferences.h‘), automated servo deadbolt driving, magnetic door-switch monitoring, automatic re-locking, anti-tamper brute-force lockouts, and immediate local telemetry via an OLED screen and buzzer alarms.

Outcomes

Expected Outcomes of the Smart Door Lock System

- **Secure authentication:** dynamic 4-to-8 digit PIN, changeable on the keypad and retained in NVS flash across power resets.
- **Brute-force protection:** 3 consecutive wrong PINs trigger a 10 s lockout with the keypad ignored and alarm chirps.
- **Automated deadbolt control:** SG90 servo locks/unlocks ($0^\circ/90^\circ$) and auto-relocks 5 s after the door closes.
- **Door-state monitoring:** reed switch detects open/closed; a 700 Hz alert sounds once the door stays open beyond 30 s.
- **Local telemetry:** OLED shows status and masked PIN feedback; red/blue LEDs and buzzer give instant lock-state cues.
- **Manual exit override:** 50 ms-debounced inside pushbutton unlocks/relocks without a PIN.
- **Remote access (in progress):** mobile app with cloud-synced lock status and toggle control, targeted by 23 Oct 2026.

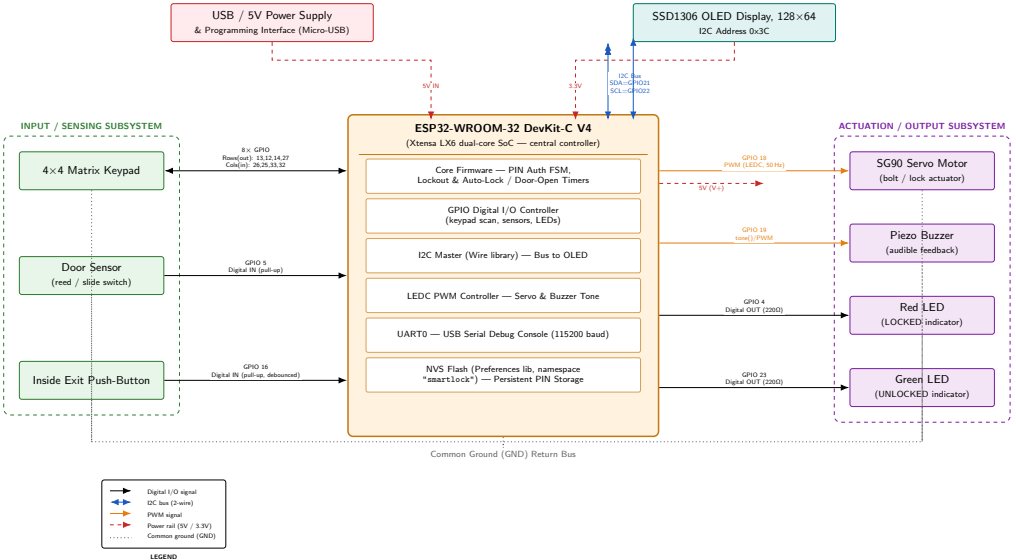
System Specifications — Core Processing & Actuation

Component	Hardware Specification	Function & Interface Mapping
Microcontroller	ESP32 DevKit V4 (3.3V Logic)	Dual-Core Xtensa LX6 @ 240MHz; handles core state machine, timed delays, and non-volatile flash memory.
Deadbolt Actuator	SG90 Micro Servo Motor	PWM-driven via GPIO 18; rotates deadbolt mechanism (0° = Locked, 90° = Unlocked).
Display Unit	0.96" SSD1306 OLED Screen	128×64 resolution over I2C bus (SDA: GPIO 21, SCL: GPIO 22); displays real-time status and masked PIN feedback.
Input Matrix	4x4 Keypad Membrane	Rows connected to GPIO 13, 12, 14, 27; Columns connected to GPIO 26, 25, 33, 32 for credential and command entry.

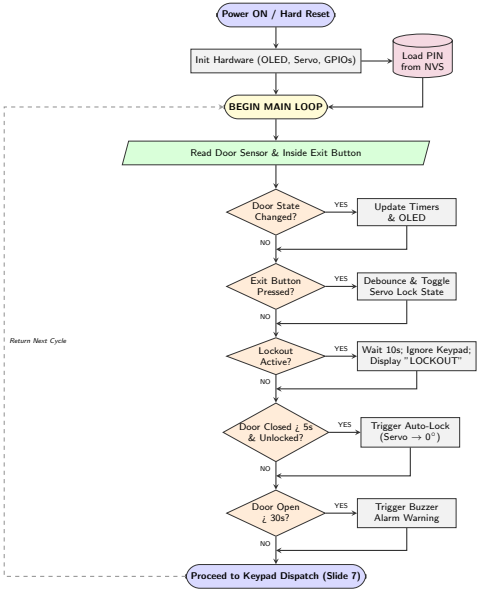
System Specifications — Sensors, Telemetry & Timers

Subsystem	Hardware / Logic Specs	Operational Constraint & Behavior
Door Switch	Magnetic Reed Contact	GPIO 5 (‘INPUT_PULLUP‘); detects physical door state (Active LOW when door frame contact is closed).
Manual Exit	Pushbutton Hardware Override	GPIO 16; 50ms software-debounced toggle allows immediate internal unlocking without PIN entry.
Telemetry	Piezo Buzzer & Status LEDs	GPIO 19 (Buzzer audio cues), GPIO 4 (Red LED - Locked), GPIO 23 (Blue LED - Unlocked).
NVS Storage	Flash Preference API	Microcontroller flash storage (‘Preferences.h‘) preserves custom passwords across reboots.
Security Timers	Finite State Machine Timers	3 failed PIN attempts $\rightarrow$ 10s system lockout; 5s auto-relock delay; 30s door-ajar alarm tone.

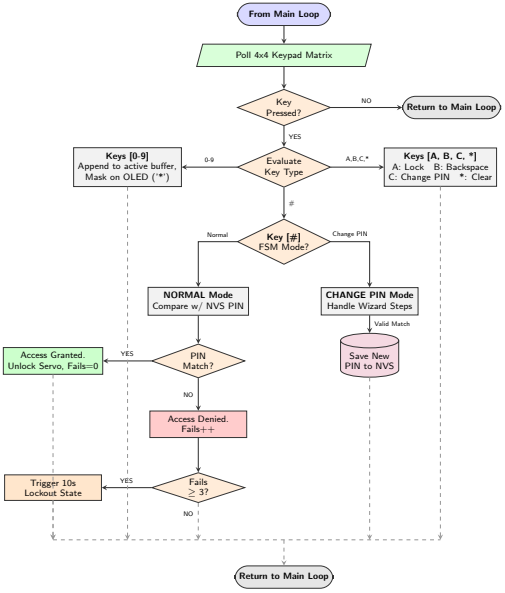
Functional Block Diagram — Hardware Architecture



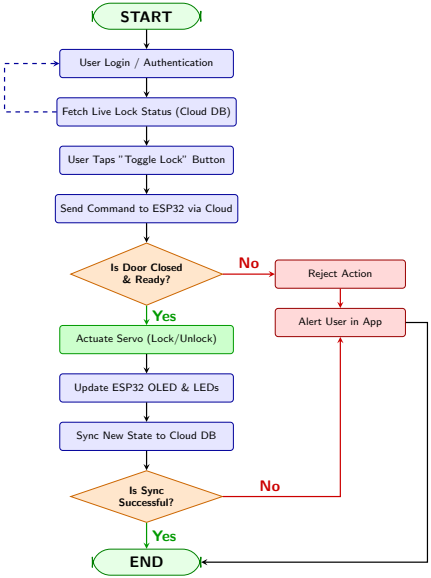
Software Architecture — Main Loop & Timers



Software Architecture — Keypad Dispatch & State Machine



Mobile App Integration Flow



Keypad Command Mapping, Edge Cases & Pin Connections

Keypad Matrix Command Mapping

- **[0 – 9]:** Appends digit to active buffer ('* * * *' masked on OLED).
- **Key [A]:** Instant lock (Normal mode only; blocked if door open).
- **Key [B]:** Backspace to remove last typed digit.
- **Key [C]:** Triggers 3-step PIN modification wizard.
- **Key [*]:** Clears input string or cancels PIN wizard.
- **Key [#]:** Submits PIN credential for state check.

System Edge Cases & Security Rules

- **3 Invalid Passwords:** Triggers 10s lockout timer; keypad ignored, alarm chirps.
- **Door Ajar ($>$ 30s):** Single 700Hz alert once door stays open past 30s (re-arms after door closes & reopens).
- **Auto-Lock (5s):** Auto-rotates deadbolt to 0° 5s after door closes.
- **Manual Hardware Exit:** 50ms-debounced pushbutton toggles lock state (unlock/re-lock), no PIN needed.

Hardware Pin Connection Table

Component	ESP32 GPIO	Target Pin / Logic
OLED Display	GPIO 21, 22	SDA, SCL (I2C Bus)
Servo Motor	GPIO 18	PWM Deadbolt Control
Piezo Buzzer	GPIO 19	Auditory Alert Signal
4x4 Keypad Rows	GPIO 13, 12, 14, 27	R1, R2, R3, R4 Matrix
4x4 Keypad Cols	GPIO 26, 25, 33, 32	C1, C2, C3, C4 Matrix
Status LEDs	GPIO 4, 23	Red (Lock) / Blue (Unlock)
Door Switch	GPIO 5	Active LOW Magnetic Sensor
Exit Button	GPIO 16	Internal Override Pushbutton

Tools Required

Hardware & Software Tools

- **Core Hardware:** ESP32 DevKit V4, SG90 Servo, SSD1306 OLED, 4x4 Keypad, Piezo Buzzer, Slide Switch, Pushbutton, LEDs,Jumper Wires, Resistors.
- **Development Software:** Wokwi Simulator, VS Code / Arduino IDE.
- **C++ Libraries:** ‘Preferences.h‘ (NVS Flash), ‘ESP32Servo‘, ‘Adafruit_SSD1306‘, ‘Keypad.h‘, ‘Wire.h‘.

Work Done

1. ESP32 Simulation Model (Wokwi)

- Full virtual circuit: ESP32 DevKit V4, SG90 servo, SSD1306 OLED, 4x4 keypad, door switch, exit pushbutton, buzzer, red/blue LEDs.
- FSM firmware: PIN entry, `NORMAL`/`CHANGE PIN` modes, 3-attempt 10 s lockout, 5 s auto-relock, 30 s door-ajar alert.
- PIN persistence across resets verified with `Preferences.h` (NVS flash); masked OLED feedback, buzzer and LED cues validated.

2. Working Real-Life Circuit

- Simulated design replicated on hardware with the same GPIO mapping (servo 18, I2C 21/22, keypad 13/12/14/27 & 26/25/33/32, door switch 5, exit button 16).
- Same firmware flashed to the physical ESP32; keypad unlock, PIN change, lockout, auto-lock, exit button and door-ajar alarm tested.
- Servo deadbolt actuation (0° locked / 90° unlocked) confirmed with the real door-state sensor.

Project Timeline

Milestones: Completed and Remaining

Step	Activity	Target	Status
1	Problem definition & literature review	–	Done
2	Component selection & pin mapping	–	Done
3	Wokwi circuit design & simulation	–	Done
4	Firmware: FSM, NVS PIN, timers, lockout	–	Done
5	Simulation testing & edge-case debugging	–	Done
6	Hardware procurement & assembly	–	Done
7	Firmware flash & real-circuit testing	–	Done
8	Mobile app: login & cloud database setup	9–13 Oct	In progress
9	App: lock-status fetch & toggle command	14–17 Oct	Pending
10	ESP32–cloud integration & state sync	18–20 Oct	Pending
11	End-to-end testing (app + hardware)	21–22 Oct	Pending
12	Final demo & documentation	**23 Oct 2026**	Pending

Reference

- **Espressif Systems (2023):** *ESP32 Non-Volatile Storage (NVS) API Reference*. Official documentation for Flash memory preference management.
- **Adafruit Industries (2022):** *SSD1306 OLED Graphics Library Documentation*. I2C communication standards for low-power embedded screens.
- **W. S. Hui and R. Sanudin (2024):** *"IoT-Based Smart Home Door Lock Security System Using ESP32"*. Faculty of Electrical and Electronic Engineering, Universiti Tun Hussein Onn Malaysia.
- **Int. Journal of Engineering Research & Technology (2026):** *"An Intelligent IoT-Based Smart Door Lock System with Remote Access and Status Display"*. IJERT Publication.
- **Wokwi Embedded Systems Platform (2026):** *Real-Time ESP32 Simulation Engine and Web-based Peripheral Interfaces*.